In [1]:

```
import pandas as pd
import numpy as np
from matplotlib import pyplot as plt
from scipy.optimize import curve_fit
```

In [2]:

```
#loading data

data = pd.read_csv('SOCR-HeightWeight.csv').set_index('Index')
data.head(5)
data.columns
```

Out[2]:

```
Index(['Height(Inches)', 'Weight(Pounds)'], dtype='object')
```

In [3]:

```
#calculating mean
data.mean()
```

Out[3]:

```
Height(Inches)     67.993114
Weight(Pounds)    127.079421
dtype: float64
```

In [4]:

```
#calculating std
data.std()
```

Out[4]:

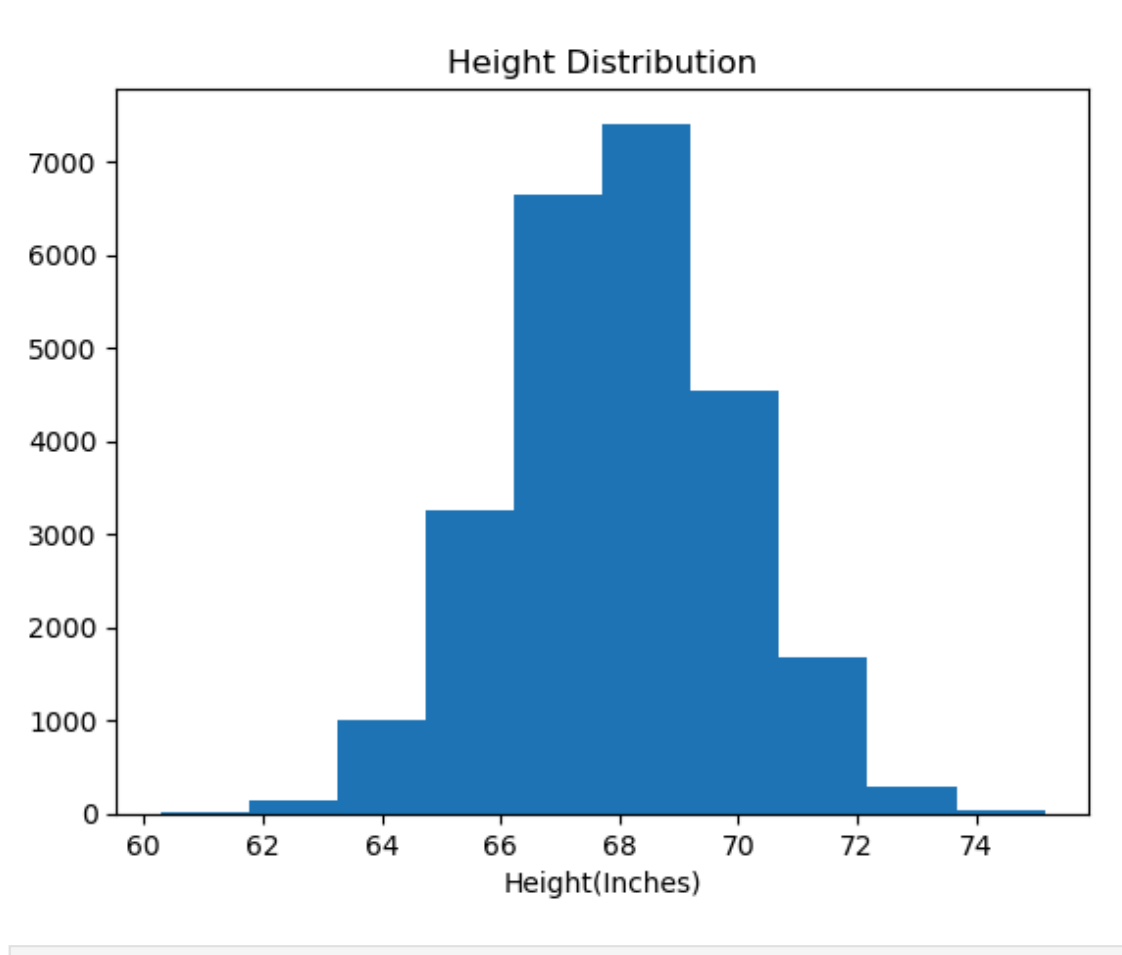
```
Height(Inches)     1.901679
Weight(Pounds)    11.660898
dtype: float64
```

In [5]:

```
plt.hist(data['Height(Inches)'])
plt.title('Height Distribution')
plt.xlabel('Height(Inches)' )
```

Out[5]:

```
Text(0.5, 0, 'Height(Inches)')
```

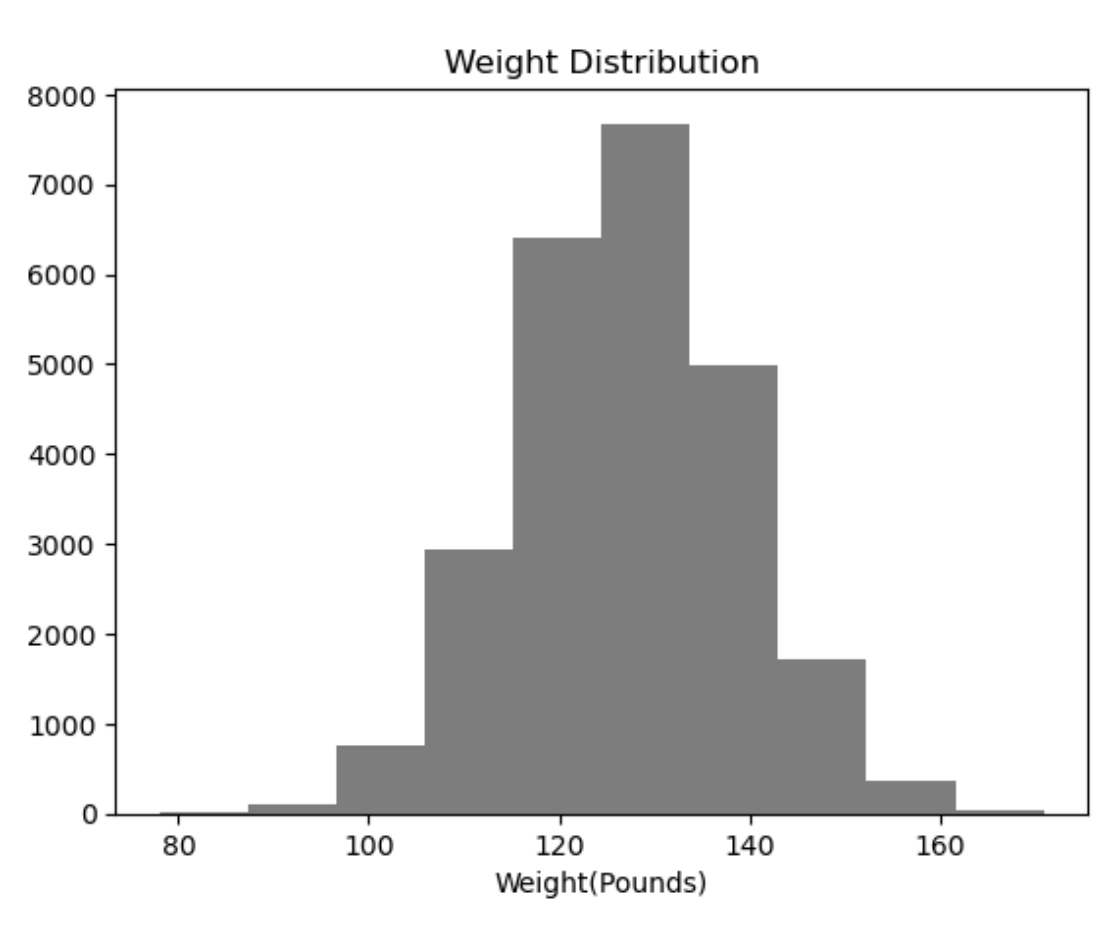


In [6]:

```
plt.hist(data['Weight(Pounds)'], color='gray')
plt.title('Weight Distribution')
plt.xlabel('Weight(Pounds)' )
```

Out[6]:

```
Text(0.5, 0, 'Weight(Pounds)')
```



$$r = \frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\sum(x_i-\bar{x})^2\sum(y_i-\bar{y})^2}}$$

In [7]:

```
#Compute the correlation between height and weight

# let height be x and weight be y
my_x = np.array(data['Height(Inches)'])
my_y = np.array(data['Weight(Pounds)'])

# a function to compute correlation between x and y
def correlation_function(x,y):
    x_bar = np.mean(my_x)
    y_bar = np.mean(my_y)
    numerator = np.sum((my_x - x_bar)*(my_y - y_bar))
    denominator = np.sqrt((np.sum((my_x - x_bar)**2)) * (np.sum((my_y - y_bar)**2)))
    r = numerator/denominator
    return r
```

In [8]:

```
correlation_function(my_x, my_y)
```

Out[8]:

```
0.5028585206028441
```

In [9]:

```
np.corrcoef(my_x,my_y)
```

Out[9]:

```
array([[1.        , 0.50285852],
       [0.50285852, 1.        ]])
```

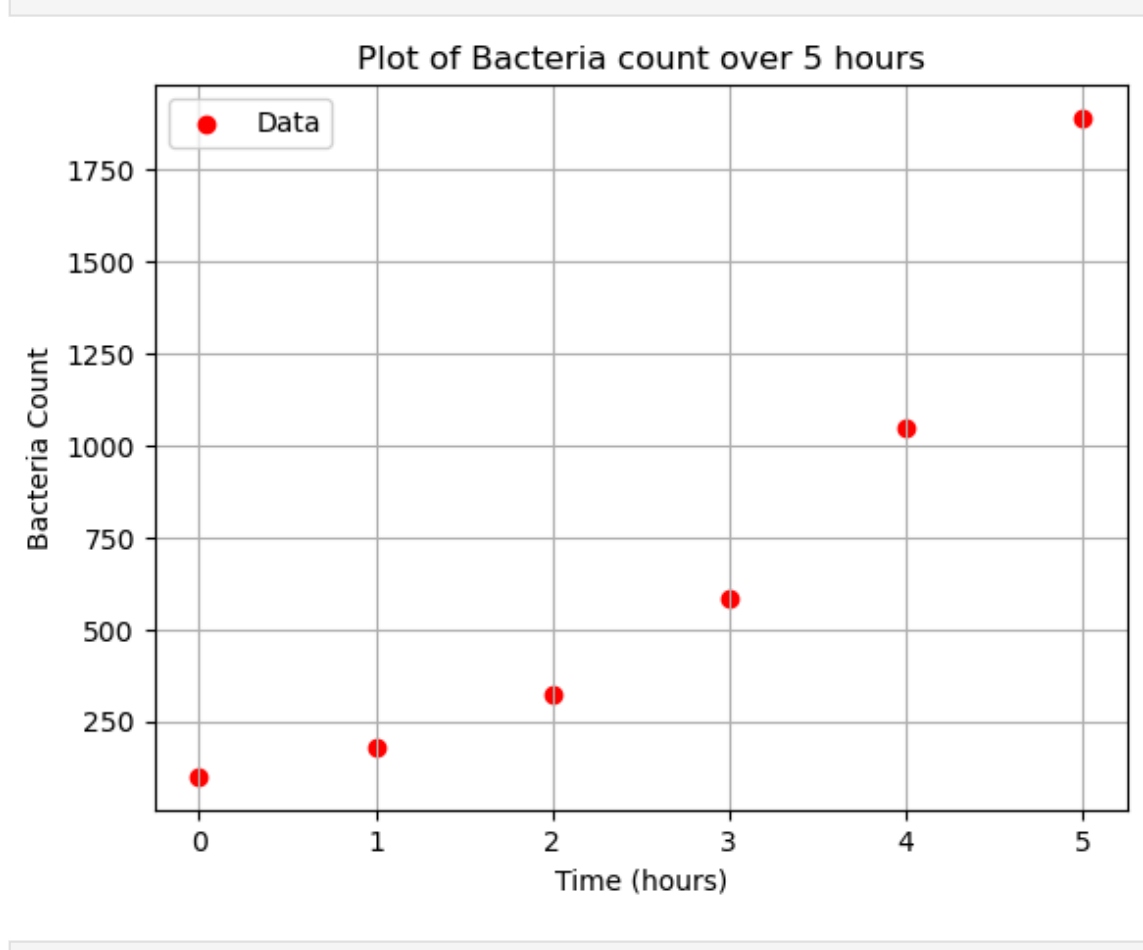
In [ ]:

In [10]:

```
# creating two lists from the data given
#let x be Time (hours) data and y be Bacteria Count
x = np.arange(0, 6)
y = np.array([100, 180, 324, 583.2, 1049.76, 1889.57], dtype=float)
```

In [11]:

```
#visualising how our data is scattered
plt.scatter(x,y, label= 'Data', c='r')
plt.xlabel('Time (hours)')
plt.ylabel('Bacteria Count')
plt.title('Plot of Bacteria count over 5 hours')
plt.legend()
plt.grid()
```



In [12]:

```
def exponential_growth(t, a,b):
    return a * np.exp(b * t)

def logistic_growth(t, K, A, B):
    return K / (1 + A * np.exp(-B * t))

def power_law(a,b,t):
    return a * (t ** b)
```

In [13]:

```
# initializing Parameters
a = 2  # example value for a
b = 0.5  # example value for b
t = x
K = 10  # Carrying capacity
```

In [14]:

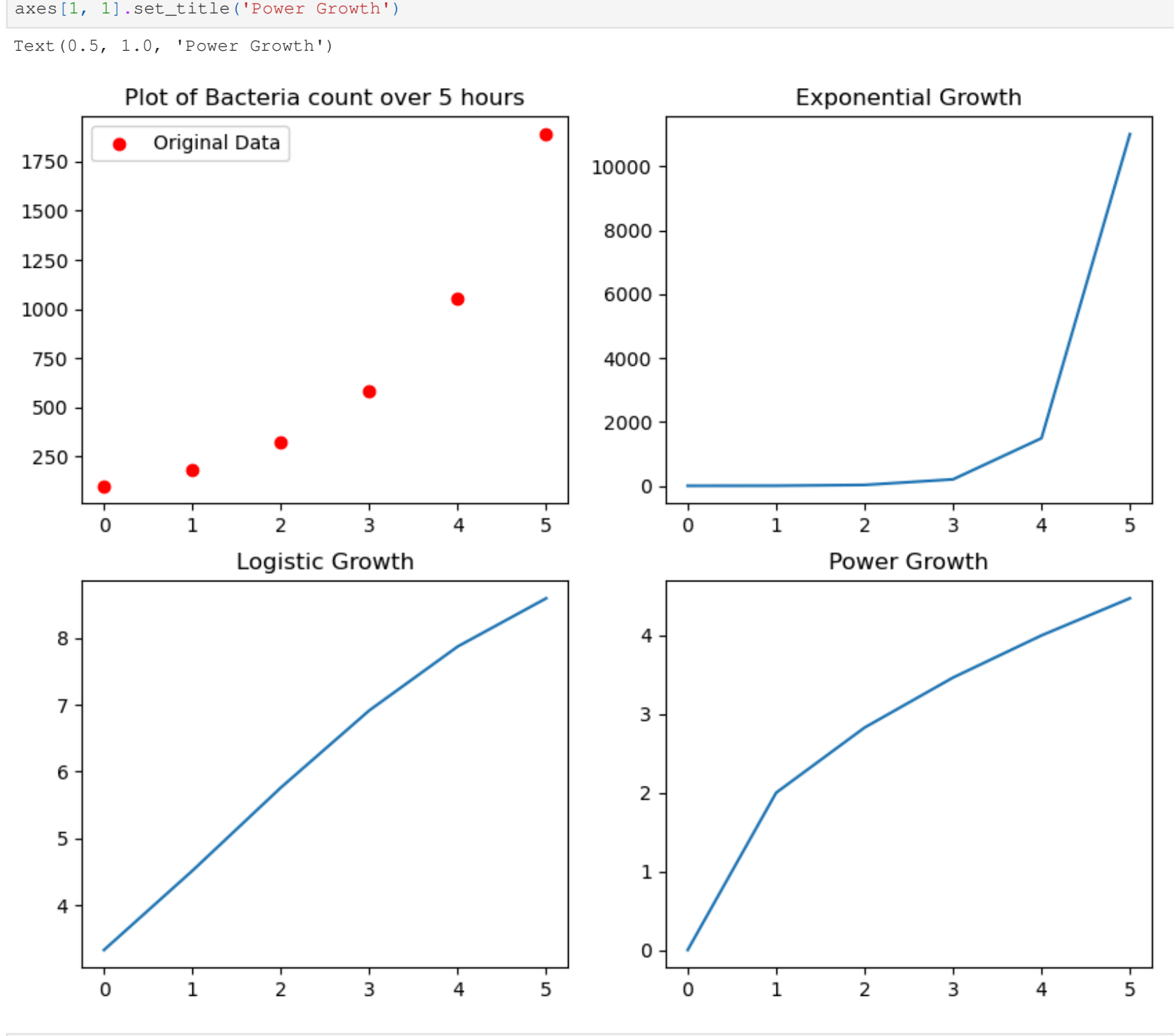
```
y_exponent = exponential_growth(a,b,t)
y_logistic = logistic_growth(t,K,a,b)
y_power = power_law(a,b,t)
```

In [15]:

```
fig, axes = plt.subplots(2, 2, figsize=(10, 8))

axes[0, 0].scatter(x,y, label= 'Original Data', c='r')
axes[0, 0].set_title('Plot of Bacteria count over 5 hours')
axes[0, 0].legend()

axes[0, 1].plot(x, y_exponent)
axes[0, 1].set_title('Exponential Growth')

axes[1, 0].plot(x, y_logistic)
axes[1, 0].set_title('Logistic Growth')

axes[1, 1].plot(x, y_power)
axes[1, 1].set_title('Power Growth')
```

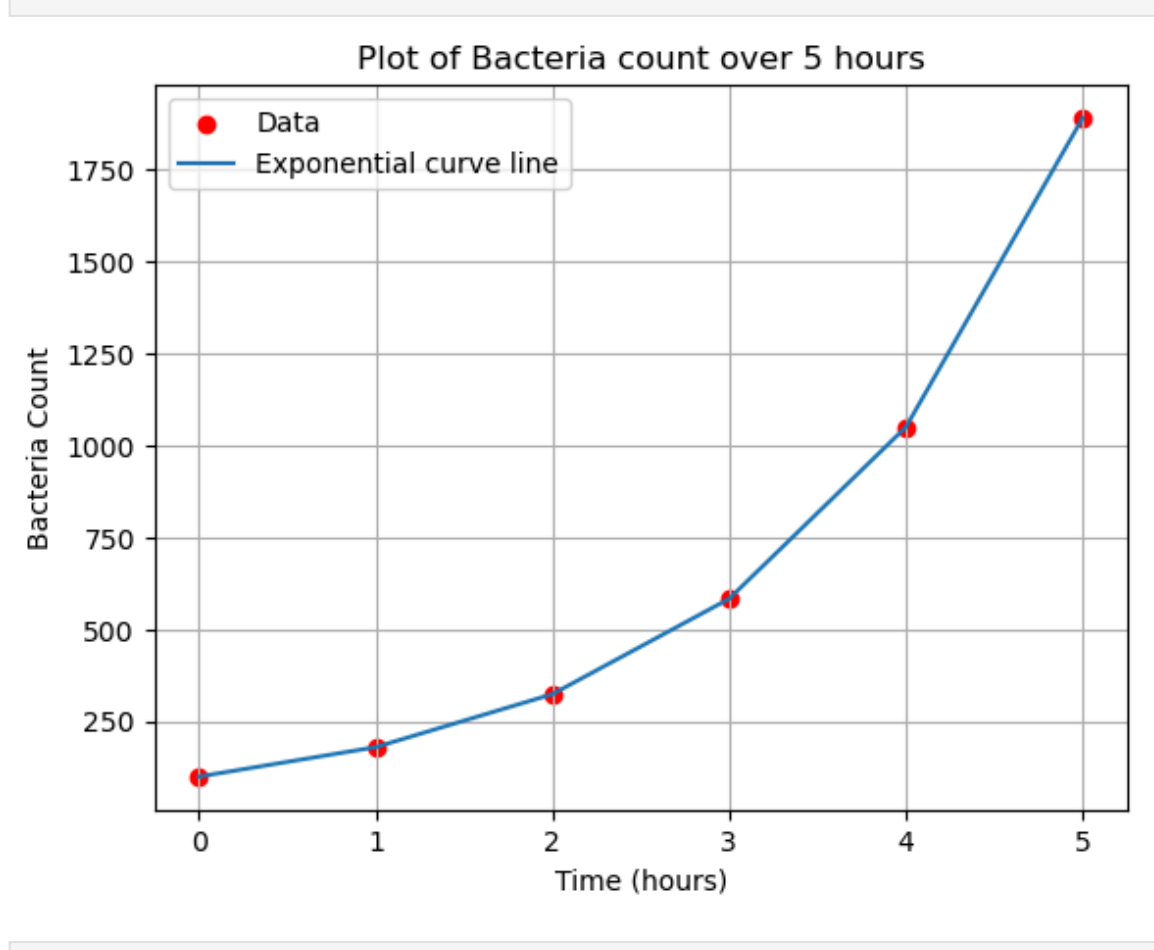
Out[15]:

```
Text(0.5, 1.0, 'Power Growth')
```



In [16]:

```
# creating two lists from the data given

# Fit the exponential curve to the data
popt, pcov = curve_fit(exponential_growth, x, y)
a_opt, b_opt = popt  # Fitted parameters a and b

# Generate the y values using the fitted curve
y_curve = exponential_growth(x, *popt)


plt.scatter(x,y, label= 'Data', color='r')
plt.plot(x,y_curve, label="Exponential curve line")
plt.xlabel('Time (hours)')
plt.ylabel('Bacteria Count')
plt.title('Plot of Bacteria count over 5 hours')
plt.legend()
plt.grid()
```



In [17]:

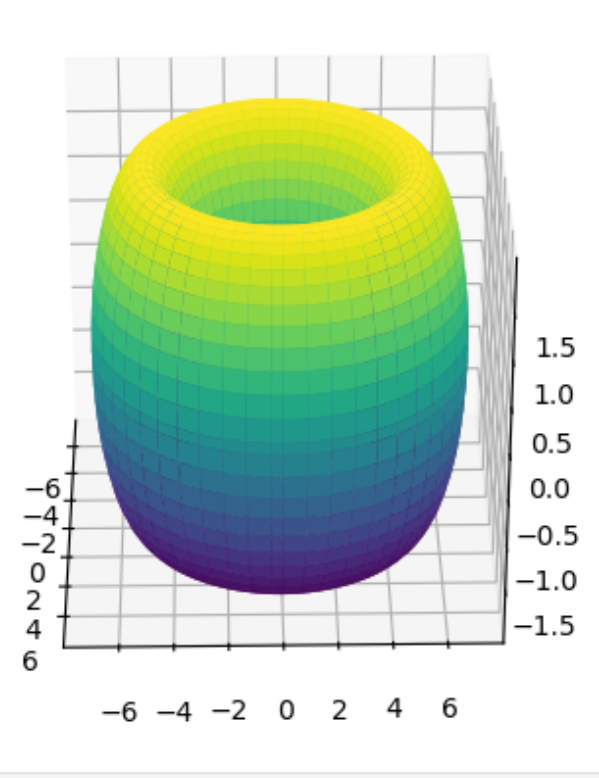
```
# Compute the residuals
residuals = y - y_curve

# Compute SS_res (sum of squared residuals)
ss_res = np.sum(residuals**2)

# Compute SS_tot (total sum of squares)
ss_tot = np.sum((y - np.mean(y))**2)

# Compute R^2
r_squared = 1 - (ss_res / ss_tot)

print(f"Coefficient of Determination (R^2): {r_squared:.4f}")
```

```
Coefficient of Determination (R^2): 1.0000
```

In [ ]:

In [119…

```
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D
from matplotlib.animation import FuncAnimation

# Parametric equations
def x_function(u, v):
    return (R + r*np.cos(v)) * np.cos(u)

def y_function(u, v):
    return (R + r*np.cos(v)) * np.sin(u)

def z_function(u, v):
    return r * np.sin(v)

# Parameters
R = 5  # Major radius
r = 2  # Minor radius
pi = np.pi
u = np.linspace(0, 2*pi, 100)
v = np.linspace(0, 2*pi, 100)

# Create a meshgrid for u, v
u, v = np.meshgrid(u, v)

# Compute x, y, z coordinates
x = x_function(u, v)
y = y_function(u, v)
z = z_function(u, v)

# Plotting
fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')

# Surface plot
ax.plot_surface(x, y, z, cmap='viridis')

# Set an equal aspect ratio
ax.set_box_aspect([1,1,1])  # Aspect ratio 1:1:1


# Define animation function
def update(frame):
    ax.view_init(elev=30, azim=frame)
    return fig,

# Create animation
ani = FuncAnimation(fig, update, frames=np.arange(0, 360, 1), interval=50)

# Save animation as a gif
ani.save('rotating_surface_plot.gif', writer='pillow')

plt.show()
```



In [ ]:

In [19]:

```
#Exploratory Analysis on Real-World Data
#Load the dataset using pandas and display all necessary information contained in the file

df = pd.read_csv('task1.txt')
df.head(5)
```

Out[19]:

	Year	Month	DayofMonth	DayOfWeek	DepTime	CRSDepTime	ArrTime	CRSArrTime	UniqueCarrier	FlightNum	...	TaxiIn	TaxiOut	Cancelled	CancellationCode	Diverted	CarrierDelay	WeatherDelay	NASDela
0	2008	1	1	2	120.0	1935	309.0	2130	9E	5746	...	3.0	18.0	0	NaN	0	339.0	0.0	0.
1	2008	1	1	2	555.0	600	826.0	835	AA	1614	...	7.0	11.0	0	NaN	0	NaN	NaN	Na
2	2008	1	1	2	600.0	600	728.0	729	YV	2883	...	7.0	16.0	0	NaN	0	NaN	NaN	Na
3	2008	1	1	2	601.0	605	727.0	750	9E	5743	...	4.0	12.0	0	NaN	0	NaN	NaN	Na
4	2008	1	1	2	601.0	600	654.0	700	AA	1157	...	5.0	10.0	0	NaN	0	NaN	NaN	Na

5 rows × 29 columns

In [20]:

```
df.isnull().sum()
```

Out[20]:

```
Year                     0
Month                    0
DayofMonth               0
DayOfWeek                0
DepTime               1413
CRSDepTime               0
ArrTime               1567
CRSArrTime               0
UniqueCarrier            0
FlightNum                0
TailNum               1104
ActualElapsedTime     1601
CRSElapsedTime          11
AirTime               1601
ArrDelay              1601
DepDelay              1413
Origin                   0
Dest                     0
Distance                 0
TaxiIn                1567
TaxiOut               1419
Cancelled                0
CancellationCode     97840
Diverted                 0
CarrierDelay         79513
WeatherDelay         79513
NASDelay             79513
SecurityDelay        79513
LateAircraftDelay    79513
dtype: int64
```

In [ ]:

In [21]:

```
df.columns
```

Out[21]:

```
Index(['Year', 'Month', 'DayofMonth', 'DayOfWeek', 'DepTime', 'CRSDepTime',
       'ArrTime', 'CRSArrTime', 'UniqueCarrier', 'FlightNum', 'TailNum',
       'ActualElapsedTime', 'CRSElapsedTime', 'AirTime', 'ArrDelay',
       'DepDelay', 'Origin', 'Dest', 'Distance', 'TaxiIn', 'TaxiOut',
       'Cancelled', 'CancellationCode', 'Diverted', 'CarrierDelay',
       'WeatherDelay', 'NASDelay', 'SecurityDelay', 'LateAircraftDelay'],
      dtype='object')
```

In [22]:

```
df['total_delay'] = df[['DepDelay', 'CarrierDelay', 'WeatherDelay', 'NASDelay', 'SecurityDelay', 'LateAircraftDelay']].sum(axis=1)
df
```

Out[22]:

	Year	Month	DayofMonth	DayOfWeek	DepTime	CRSDepTime	ArrTime	CRSArrTime	UniqueCarrier	FlightNum	...	TaxiOut	Cancelled	CancellationCode	Diverted	CarrierDelay	WeatherDelay	NASDelay
0	2008	1	1	2	120.0	1935	309.0	2130	9E	5746	...	18.0	0	NaN	0	339.0	0.0	0.0
1	2008	1	1	2	555.0	600	826.0	835	AA	1614	...	11.0	0	NaN	0	NaN	NaN	NaN
2	2008	1	1	2	600.0	600	728.0	729	YV	2883	...	16.0	0	NaN	0	NaN	NaN	NaN
3	2008	1	1	2	601.0	605	727.0	750	9E	5743	...	12.0	0	NaN	0	NaN	NaN	NaN
4	2008	1	1	2	601.0	600	654.0	700	AA	1157	...	10.0	0	NaN	0	NaN	NaN	NaN
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
99255	2008	12	31	3	2109.0	2100	2158.0	2157	CO	1533	...	18.0	0	NaN	0	NaN	NaN	NaN
99256	2008	12	31	3	2124.0	2123	2246.0	2300	9E	5805	...	22.0	0	NaN	0	NaN	NaN	NaN
99257	2008	12	31	3	2327.0	2325	9.0	20	AA	1532	...	7.0	0	NaN	0	NaN	NaN	NaN
99258	2008	12	31	3	NaN	1220	NaN	1711	B6	1264	...	NaN	1	B	0	NaN	NaN	NaN
99259	2008	12	31	3	NaN	1300	NaN	1617	B6	1065	...	NaN	1	B	0	NaN	NaN	NaN

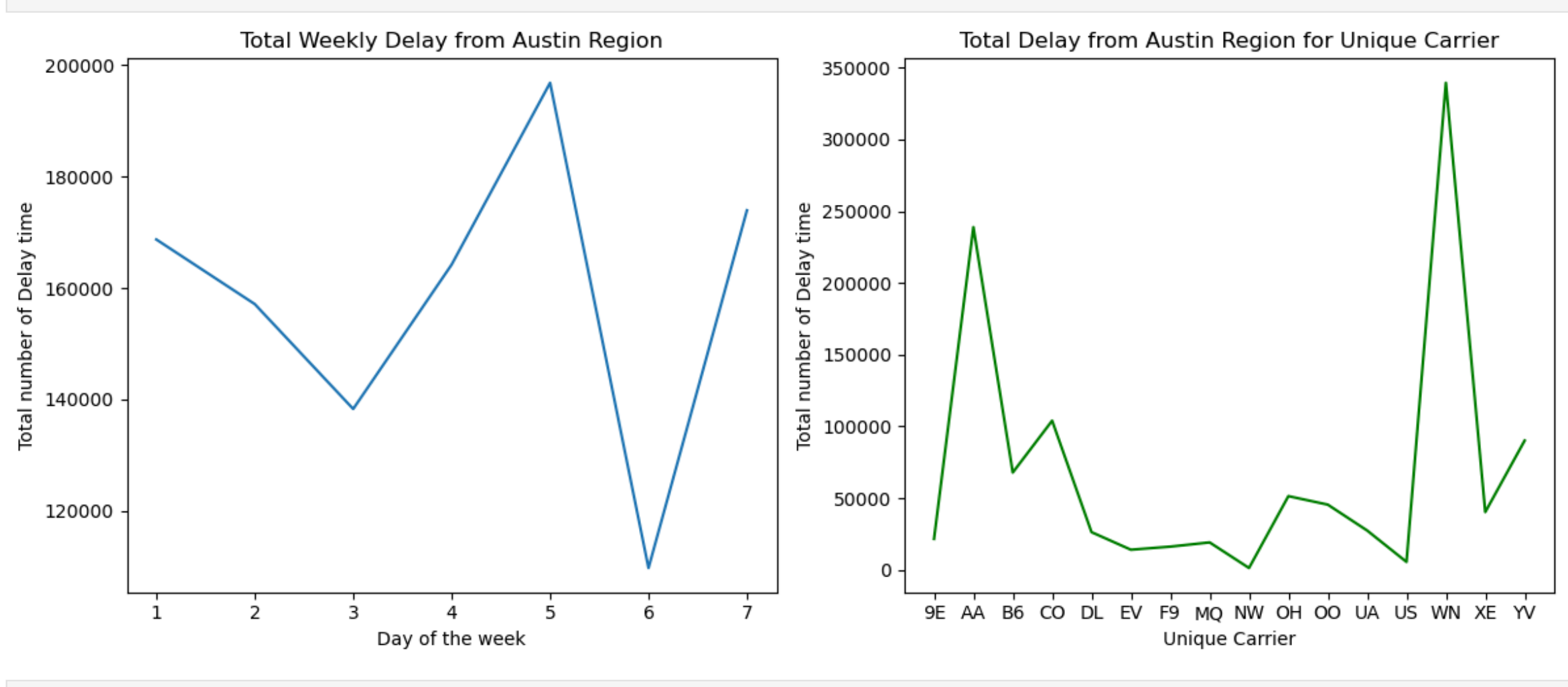
99260 rows × 30 columns

In [23]:

```
# time of the day, possible delays, change with airlines
#'DayOfWeek'vs 'DepDelay' later we do 'DayOfWeek' vs 'CarrierDelay'
austin_region = df[df['Dest'] == 'AUS']

general_daily_delay = austin_region[['DayOfWeek','total_delay']].groupby('DayOfWeek').sum()
carrier_daily_delay = austin_region[['UniqueCarrier','total_delay']].groupby('UniqueCarrier').sum()
```

In [27]:

```
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 5))

ax1.plot(general_daily_delay['total_delay'])
ax1.set_title('Total Weekly Delay from Austin Region')
ax1.set_xlabel("Day of the week")
ax1.set_ylabel("Total number of Delay time")

ax2.plot(carrier_daily_delay['total_delay'], c='g')
ax2.set_title('Total Delay from Austin Region for Unique Carrier')
ax2.set_xlabel("Unique Carrier")
ax2.set_ylabel("Total number of Delay time")

# Adjust layout to prevent overlap
plt.tight_layout()
```



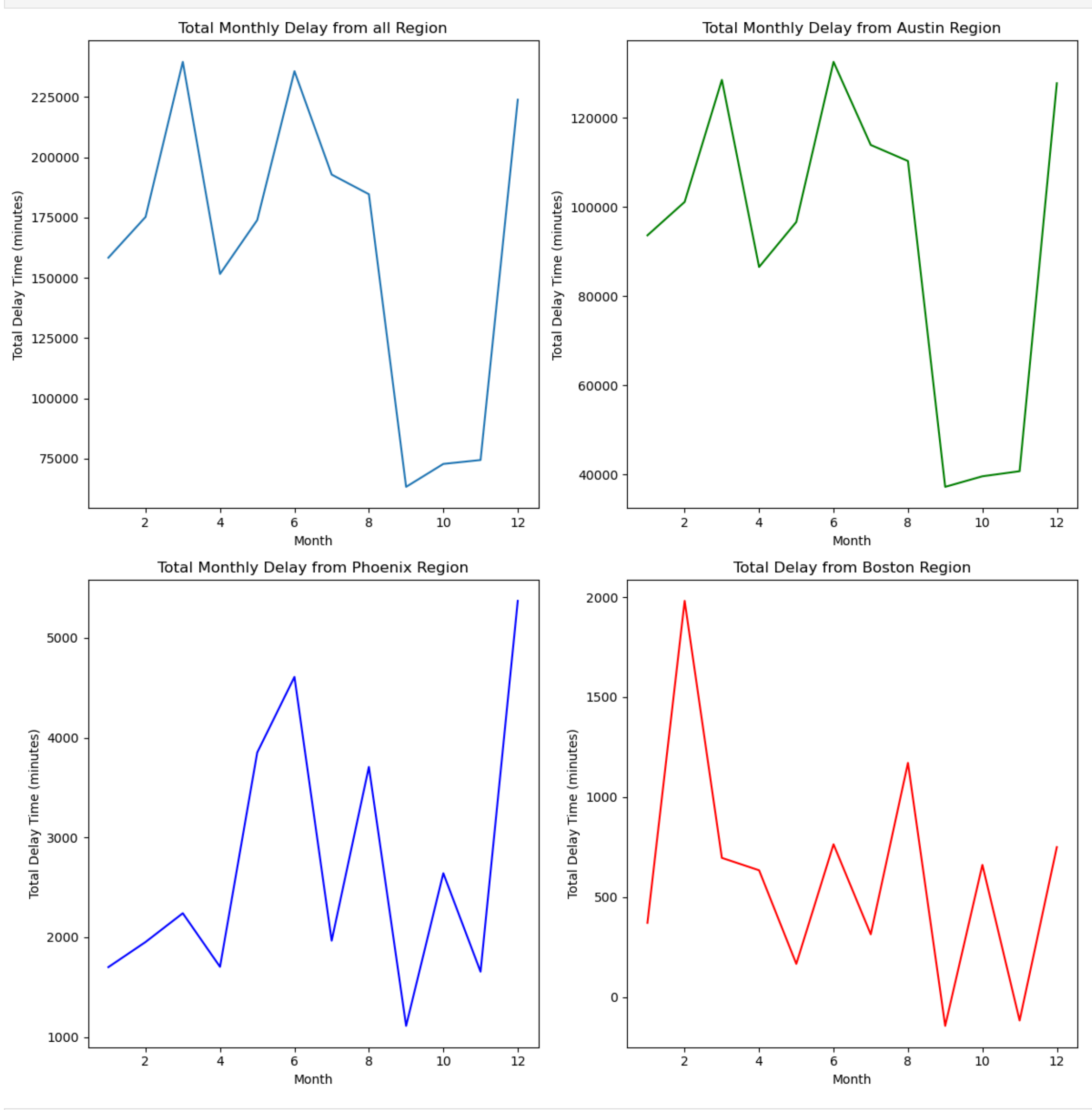
In [ ]:

In [28]:

```
#Investigate what time of the year it is more suited to fly so as to have the delays minimum and
#does the destination affect this? You can lay insights on some popular destinations for the task.

general_monthly_delay = df[['Month','total_delay']].groupby('Month').sum()
austin_monthly_delay = austin_region[['Month','total_delay' ]].groupby('Month').sum()
PHX_region = df[df['Dest'] == 'PHX']
PHX_monthly_delay = PHX_region[['Month','total_delay' ]].groupby('Month').sum()
BOS_region = df[df['Dest'] == 'BOS']
BOS_monthly_delay = BOS_region[['Month','total_delay' ]].groupby('Month').sum()
```

In [29]:

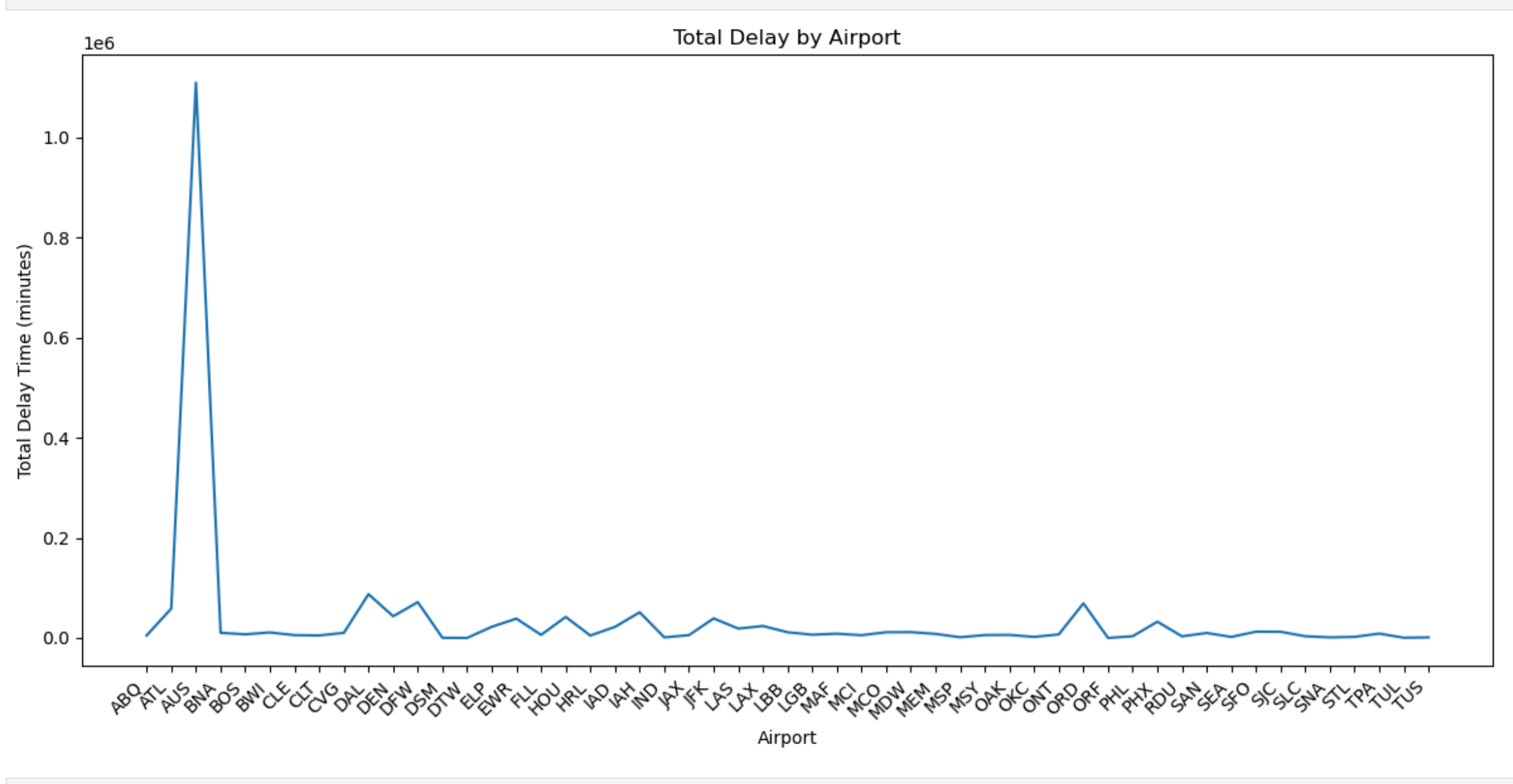
```
fig, axes = plt.subplots(2, 2, figsize=(12, 12))

# Flatten the 2D array of axes for easier indexing
(ax1, ax2), (ax3, ax4) = axes

# Plot total weekly delay from Austin region
ax1.plot(general_monthly_delay['total_delay'])
ax1.set_title('Total Monthly Delay from all Region')
ax1.set_xlabel("Month")
ax1.set_ylabel("Total Delay Time (minutes)")

# Plot total delay from Austin region for a unique carrier
ax2.plot(austin_monthly_delay['total_delay'], c='g')
ax2.set_title('Total Monthly Delay from Austin Region')
ax2.set_xlabel("Month")
ax2.set_ylabel("Total Delay Time (minutes)")

# Plot total delay from Austin region to Phoenix
ax3.plot(PHX_monthly_delay['total_delay'], c='b')
ax3.set_title('Total Monthly Delay from Phoenix Region')
ax3.set_xlabel("Month")
ax3.set_ylabel("Total Delay Time (minutes)")

# Plot total delay from Austin region to Boston
ax4.plot(BOS_monthly_delay['total_delay'], c='r')
ax4.set_title('Total Delay from Boston Region')
ax4.set_xlabel("Month")
ax4.set_ylabel("Total Delay Time (minutes)")

# Adjust layout to prevent overlap
plt.tight_layout()
```



In [ ]:

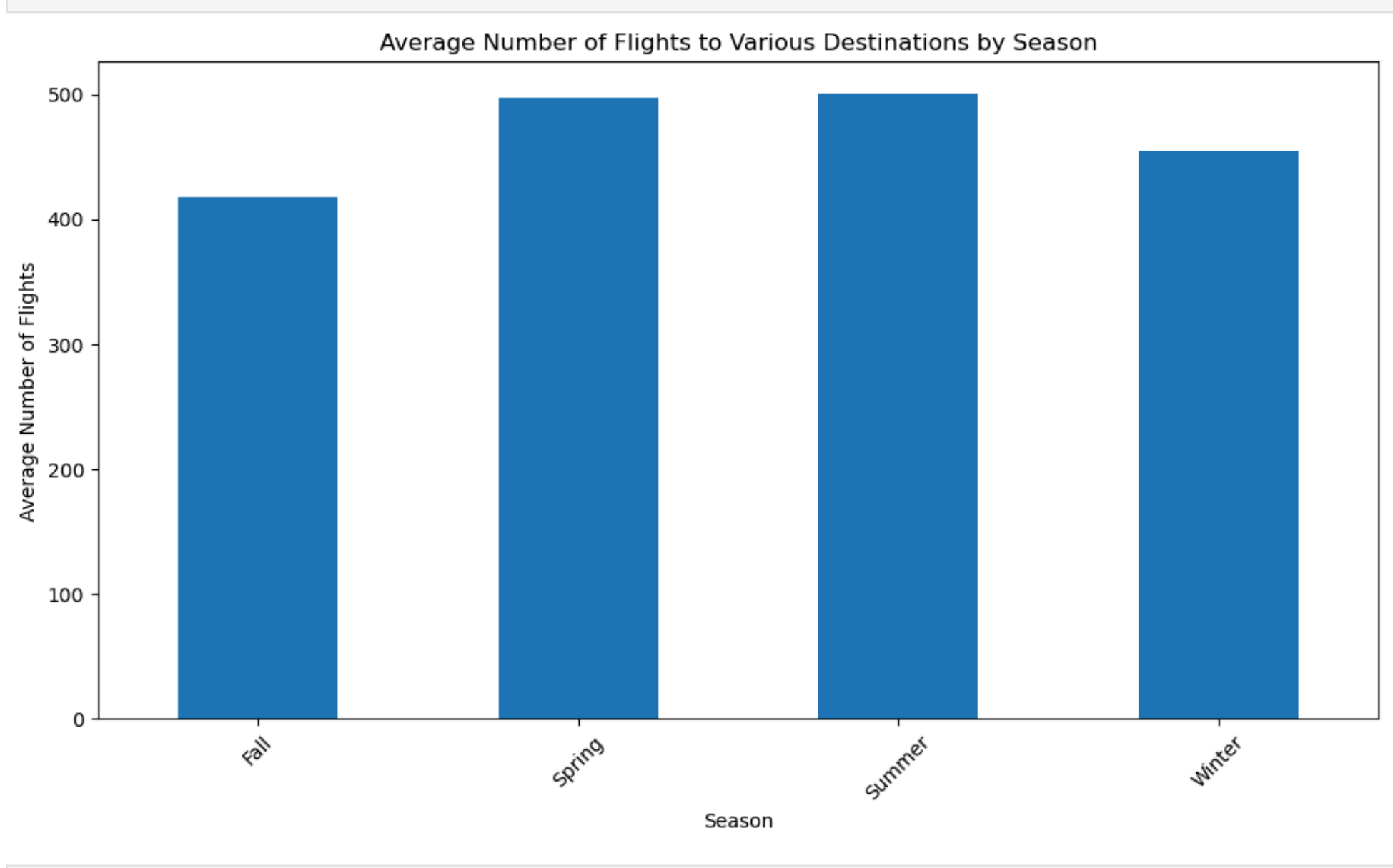
In [30]:

```
airport_delay = df[['Dest', 'total_delay']].groupby('Dest').sum()
# Create a plot with the specified figure size
plt.figure(figsize=(12, 6))

# Plot the total delay
plt.plot(airport_delay['total_delay'])

# Set title and labels
plt.title('Total Delay by Airport')
plt.xlabel("Airport")
plt.ylabel("Total Delay Time (minutes)")

# Rotate x-axis labels to make airport names visible
plt.xticks(ticks=range(len(airport_delay.index)), labels=airport_delay.index, rotation=45, ha='right')

# Adjust layout to prevent overlap
plt.tight_layout()
```



In [33]:

```
# Define seasons
def get_season(month):
    if month in [12, 1, 2]:
        return 'Winter'
    elif month in [3, 4, 5]:
        return 'Spring'
    elif month in [6, 7, 8]:
        return 'Summer'
    else:
        return 'Fall'

# Create a new column for seasons
df['Season'] = df['Month'].apply(get_season)

# Count flights by season and destination
seasonal_flight_counts = df.groupby(['Season', 'Dest']).size().unstack(fill_value=0)

# Calculate the average number of flights per destination for each season
seasonal_average = seasonal_flight_counts.mean(axis=1)

# Create a bar plot for average flights by season
plt.figure(figsize=(10, 6))
seasonal_average.plot(kind='bar')

# Set title and labels
plt.title('Average Number of Flights to Various Destinations by Season')
plt.xlabel('Season')
plt.ylabel('Average Number of Flights')
plt.xticks(rotation=45)

# Show the plot
plt.tight_layout()
plt.show()
```



In [68]:

```
sport_data = pd.read_csv('task2.txt')
sport_data.head(5)
```

Out[68]:

	id	name	sex	age	height	weight	team	noc	games	year	season	city	sport	event	medal
0	16	Juhamatti Tapio Aaltonen	M	28	184	85.0	Finland	FIN	2014 Winter	2014	Winter	Sochi	Ice Hockey	Ice Hockey Men's Ice Hockey	Bronze
1	17	Paavo Johannes Aaltonen	M	28	175	64.0	Finland	FIN	1948 Summer	1948	Summer	London	Gymnastics	Gymnastics Men's Individual All-Around	Bronze
2	17	Paavo Johannes Aaltonen	M	28	175	64.0	Finland	FIN	1948 Summer	1948	Summer	London	Gymnastics	Gymnastics Men's Team All-Around	Gold
3	17	Paavo Johannes Aaltonen	M	28	175	64.0	Finland	FIN	1948 Summer	1948	Summer	London	Gymnastics	Gymnastics Men's Horse Vault	Gold
4	17	Paavo Johannes Aaltonen	M	28	175	64.0	Finland	FIN	1948 Summer	1948	Summer	London	Gymnastics	Gymnastics Men's Pommelled Horse	Gold

In [ ]:

In [70]:

```
female_athletics_data = sport_data[
    (sport_data['sport'] == 'Athletics') &
    (sport_data['sex'] == 'F') &
    (sport_data['event']== "Athletics Women's 100 metres")]

height_95th_percentile = female_athletics_data["height"].quantile(0.95)
print(f"95th percentile of heights for female Athletics competitors: {height_95th_percentile} cm")
```

```
95th percentile of heights for female Athletics competitors: 179.6 cm
```

In [91]:

```
#Find the single woman’s event that depicts the highest variability in the height of the competitor
#across the entire history of Olympics. Use the standard deviation as the yardstick for this.
female_events = sport_data[sport_data['sex'] == 'F']
female_events_std_dev = female_events[['event', 'height']].groupby('event').std()
event_with_highest_std_dev = female_events_std_dev.idxmax()
print(event_with_highest_std_dev[0])
```

```
Rowing Women's Coxed Fours
```

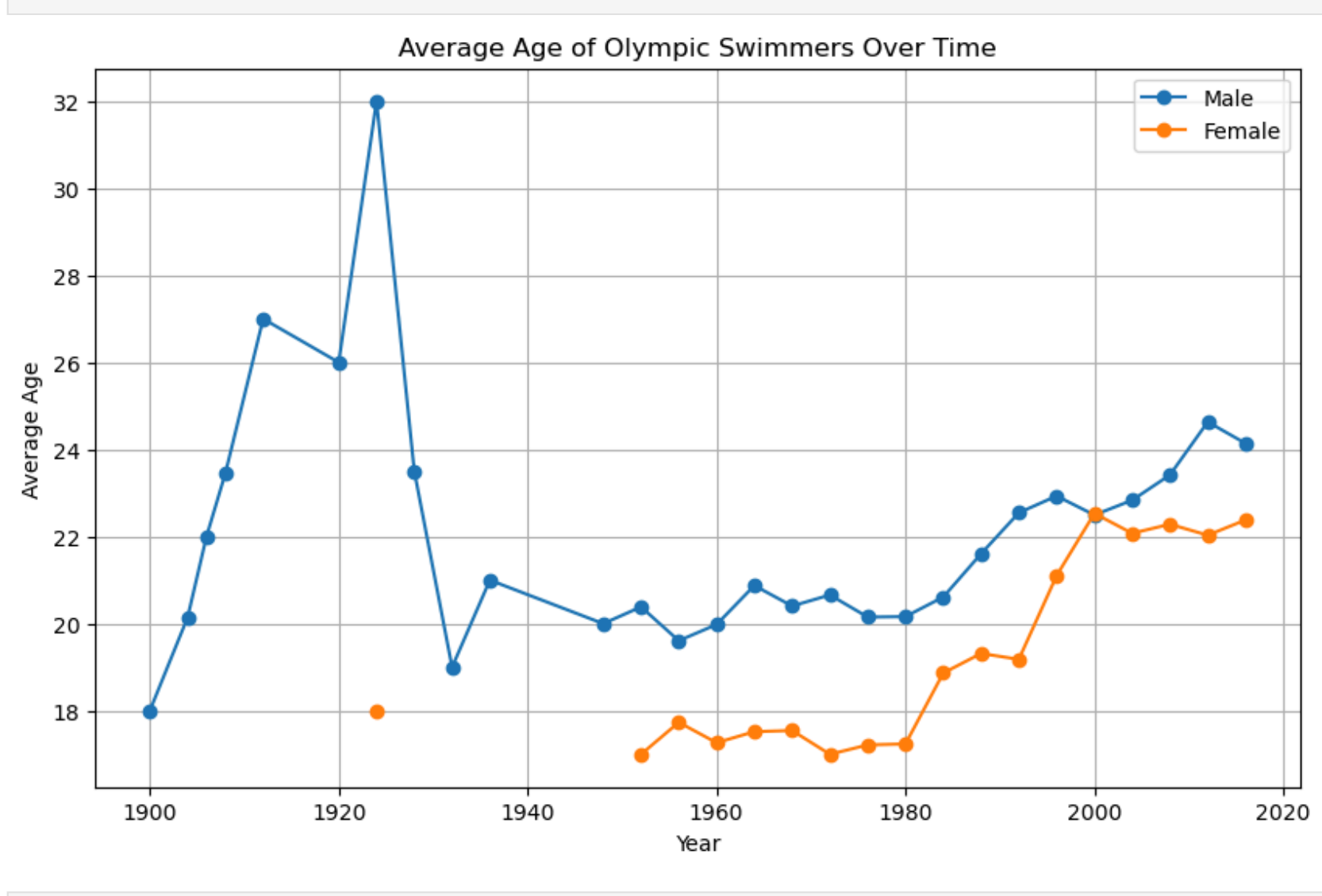
In [ ]:

In [115…

```
#'age','sex','year'sport
#average age of swimmers
swimmer_data = sport_data[sport_data['sport'] == "Swimming"]
average_age_by_year_gender = swimmer_data.groupby(["year", "sex"])["age"].mean().unstack()
```

In [114…

```
plt.figure(figsize=(10, 6))
plt.plot(average_age_by_year_gender.index, average_age_by_year_gender["M"], label="Male", marker='o')
plt.plot(average_age_by_year_gender.index, average_age_by_year_gender["F"], label="Female", marker='o')

plt.xlabel("Year")
plt.ylabel("Average Age")
plt.title("Average Age of Olympic Swimmers Over Time")
plt.legend()
plt.grid(True)

plt.show()
```



In [ ]: